

4.13.58

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Question 4.13.58 Show that all chords of the curve $3x^2 - y^2 - 2x + 4y = 0$, which subtend a right angle at the origin, pass through a fixed point. Find the coordinates of the point.

Solution: The equation of the given curve is $3x^2 - y^2 - 2x + 4y = 0$

Let $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$.

The equation of the curve can be written in the matrix form $\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0$ as

$$\mathbf{x}^T \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}^T \mathbf{x} + 0 = 0. \quad (0.1)$$

Here, the matrix of the quadratic part is

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \quad (0.2)$$

, the vector for the linear part is

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad (0.3)$$

and the constant term is $f = 0$.

Let the equation of a chord PQ of the curve be $lx + my = 1$.

In vector form, this is:

$$\mathbf{n}^T \mathbf{x} = 1, \quad (0.4)$$

$$\text{where } \mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \quad (0.5)$$

The combined equation of the pair of lines joining the origin to the points of intersection, P and Q, is obtained by homogenizing equation (0.1) using equation (0.4).

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2(\mathbf{u}^T \mathbf{x})(1) = 0 \quad (0.6)$$

Substituting $\mathbf{n}^T \mathbf{x}$ for 1:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2(\mathbf{u}^T \mathbf{x})(\mathbf{n}^T \mathbf{x}) = 0 \quad (0.7)$$

This can be written as a single quadratic form:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + \mathbf{x}^T (\mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T) \mathbf{x} = 0 \quad (0.8)$$

$$\mathbf{x}^T (\mathbf{V} + \mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T) \mathbf{x} = 0 \quad (0.9)$$

This is the equation of the pair of lines OP and OQ. Let

$$\mathbf{W} = \mathbf{V} + \mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T. \quad (0.10)$$

The lines represented by the homogeneous quadratic equation $\mathbf{x}^T \mathbf{W} \mathbf{x} = 0$ are perpendicular if and only if the trace of the matrix $\mathbf{W}$ is zero. $\text{trace}(\mathbf{W}) = 0$

A homogeneous quadratic equation represents a pair of straight lines passing through the origin. Such an equation can be factored into a product of two linear forms, each representing one of the lines.

Factoring the Quadratic Form: The equation $\mathbf{x}^T \mathbf{W} \mathbf{x} = 0$ represents two lines passing through the origin. Let the normal vectors to these lines be $\mathbf{n}_1$ and $\mathbf{n}_2$. The equation for the pair of lines can then be written as the product of the individual line equations:

$$(\mathbf{n}_1^T \mathbf{x})(\mathbf{n}_2^T \mathbf{x}) = 0$$

This product of linear forms corresponds to the quadratic form $\mathbf{x}^T \mathbf{W} \mathbf{x} = 0$, where the symmetric matrix $\mathbf{W}$ is given by:

$$\mathbf{W} = k(\mathbf{n}_1 \mathbf{n}_2^T + \mathbf{n}_2 \mathbf{n}_1^T)$$

for some non-zero scalar constant k . We can let $k = 1/2$ for simplicity without loss of generality. **Condition for Perpendicular Lines:** Two lines are **perpendicular** if and only if their respective normal vectors are orthogonal. The condition for the orthogonality of the normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ is that their scalar product is zero:

$$\mathbf{n}_1^T \mathbf{n}_2 = 0$$

Connecting to the Trace: Now, let's compute the trace of the matrix $\mathbf{W}$. Using the properties of the trace operator:

- $\text{trace}(\mathbf{A} + \mathbf{B}) = \text{trace}(\mathbf{A}) + \text{trace}(\mathbf{B})$
- $\text{trace}(\mathbf{a} \mathbf{b}^T) = \mathbf{b}^T \mathbf{a}$

We have:

$$\begin{aligned}
 \text{trace}(\mathbf{W}) &= \text{trace}\left(\frac{1}{2}(\mathbf{n}_1\mathbf{n}_2^\top + \mathbf{n}_2\mathbf{n}_1^\top)\right) \\
 &= \frac{1}{2}[\text{trace}(\mathbf{n}_1\mathbf{n}_2^\top) + \text{trace}(\mathbf{n}_2\mathbf{n}_1^\top)] && \text{(by linearity of trace)} \\
 &= \frac{1}{2}(\mathbf{n}_2^\top\mathbf{n}_1 + \mathbf{n}_1^\top\mathbf{n}_2) && \text{(using the second property)} \\
 &= \frac{1}{2}(2 \cdot \mathbf{n}_1^\top\mathbf{n}_2) && \text{(since scalar product is commutative)} \\
 &= \mathbf{n}_1^\top\mathbf{n}_2
 \end{aligned}$$

We have shown that $\text{trace}(\mathbf{W})$ is equal to the scalar product $\mathbf{n}_1^\top\mathbf{n}_2$. Therefore, the condition for the lines to be perpendicular ($\mathbf{n}_1^\top\mathbf{n}_2 = 0$) is equivalent to the condition that the trace of the matrix $\mathbf{W}$ is zero.

$$\text{trace}(\mathbf{W}) = 0 \iff \mathbf{n}_1^\top\mathbf{n}_2 = 0$$

This completes the proof.

$$\text{trace}(\mathbf{W}) = \text{trace}(\mathbf{V}) + \text{trace}(\mathbf{u}\mathbf{n}^\top) + \text{trace}(\mathbf{n}\mathbf{u}^\top) \quad (0.11)$$

$$\text{trace}(\mathbf{W}) = \text{trace}(\mathbf{V}) + \mathbf{n}^\top\mathbf{u} + \mathbf{u}^\top\mathbf{n} = \text{trace}(\mathbf{V}) + 2\mathbf{u}^\top\mathbf{n} \quad (0.12)$$

The condition for the chord to subtend a right angle at the origin is

$$\text{trace}(\mathbf{V}) + 2\mathbf{u}^\top\mathbf{n} = 0 \quad (0.13)$$

From the problem statement:

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \quad (0.14)$$

$$\implies \text{trace}(\mathbf{V}) = 3 + (-1) = 2 \quad (0.15)$$

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad (0.16)$$

$$\mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \quad (0.17)$$

Substituting these values into equation (0.13)

$$2 + 2\begin{pmatrix} -1 & 2 \end{pmatrix} \begin{pmatrix} l \\ m \end{pmatrix} = 0 \quad (0.18)$$

$$2 + 2(-l + 2m) = 0 \quad (0.19)$$

$$l - 2m = 1 \quad (0.20)$$

This is the condition that the coefficients of the chord $lx + my = 1$ must satisfy. We want to show that this line passes through a fixed point (x_0, y_0) . If it does, then the coordinates of this point must satisfy the equation of the line for any l and m that satisfy condition (0.23).

$$lx_0 + my_0 = 1 \quad (0.21)$$

From condition (0.14), we express l in terms of m : $l = 1 + 2m$. Substituting this into equation (0.24):

$$(1 + 2m)x_0 + my_0 = 1 \quad (0.22)$$

$$(x_0 - 1) + m(2x_0 + y_0) = 0 \quad (0.23)$$

Since this equation must hold for all possible values of m (which corresponds to all possible chords subtending a right angle), the coefficients of the powers of m must be individually zero.

$$x_0 - 1 = 0 \quad (0.24)$$

$$\implies x_0 = 1 \quad (0.25)$$

$$2x_0 + y_0 = 0 \quad (0.26)$$

$$\implies 2(1) + y_0 = 0 \quad (0.27)$$

$$\implies y_0 = -2 \quad (0.28)$$

Therefore, all such chords pass through the fixed point with coordinates $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

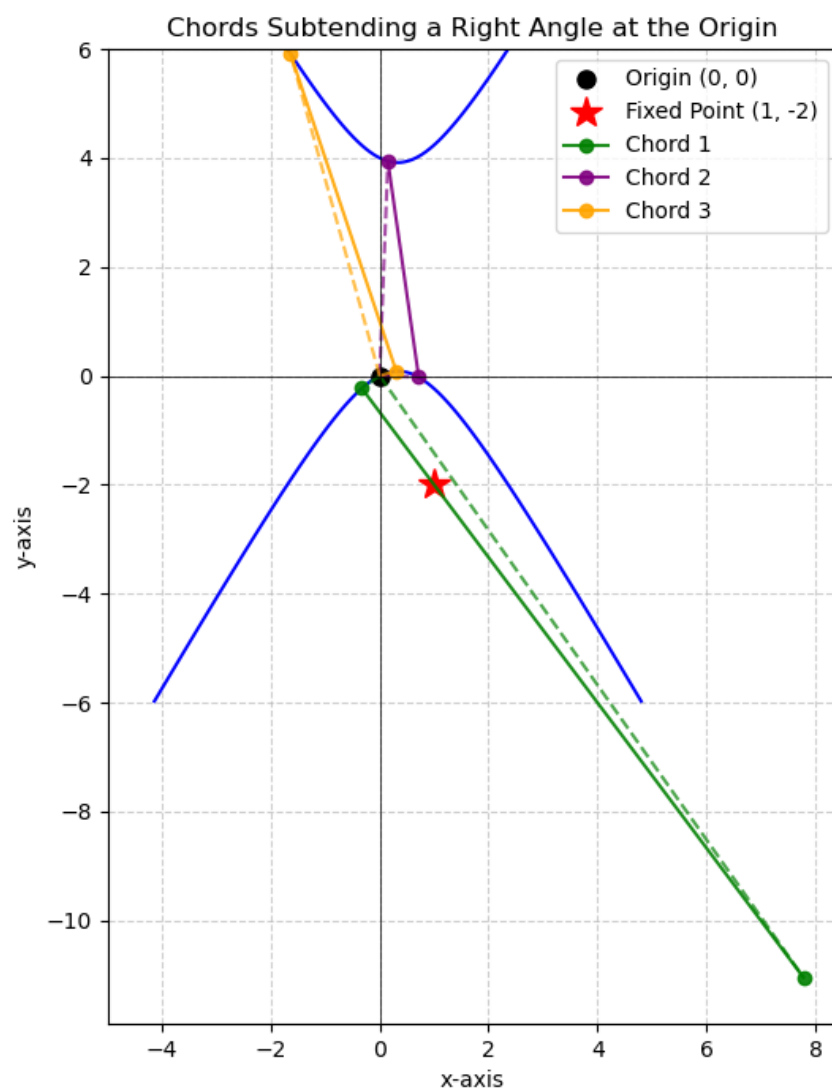


Fig. 0.1